

1. A power company wants to predict the electricity demand in different regions based on past usage, temperature, and time of year. What type of machine learning approach would be suitable? Outline the steps to build the model.

A: Supervised Learning – Regression (because the output is a continuous value: predict the electricity demand).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the dataset into training and testing sets (e.g., 80% train, 20% test).
5. Choose a regression model and train the model using the training data.
6. Evaluate model performance using metrics like R^2 score on the test data.
7. Fine-tune the model with GridSearch CV and then deploy the best model

2. An e-commerce platform wants to identify and filter out fake product reviews based on user behavior, review patterns, and sentiment analysis. What type of machine learning approach would be suitable? Outline the steps to build the model.

A: Supervised Learning – Classification (because the output is a category: fake or genuine review).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the data into training and testing sets.
5. Choose a classification model and train it.
6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
7. Fine-tune using GridSearchCV and deploy the best model

3. A city traffic department wants to estimate traffic congestion levels based on GPS data, road construction reports, and weather conditions. What type of machine learning approach would be suitable? Outline the steps to build the model.

A: Supervised Learning – Regression (because the output is a continuous value: estimate traffic congestion levels).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the dataset into training and testing sets (e.g., 80% train, 20% test).
5. Choose a regression model and train the model using the training data.
6. Evaluate model performance using metrics like R^2 score on the test data.
7. Fine-tune the model with GridSearch CV and then deploy the best model

4. A bank wants to determine whether a loan applicant should be approved or rejected based on income, credit history, and previous loan repayment behavior. What type of machine learning approach would be suitable? Outline the steps to build the model.

A: Supervised Learning – Classification (because the output is a category: loan applicant should be approved or rejected).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
 2. Encode categorical using OneHotEncoder or LabelEncoder.
 3. Scale numerical features using StandardScaler.
 4. Split the data into training and testing sets.
 5. Choose a classification model and train it.
 6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
 7. Fine-tune using GridSearchCV and deploy the best model
5. A factory wants to automatically detect defective products using images from a quality control camera. What type of machine learning approach would be suitable? Outline the steps to build the model.
- A:** Supervised Learning – Classification (because the output is a category: defective or not defective, based on labeled images).
- Step by Step Logic:**
1. Use data given, Clean the data by handling missing values and outliers.
 2. Encode categorical using OneHotEncoder or LabelEncoder.
 3. Scale numerical features using StandardScaler.
 4. Split the data into training and testing sets.
 5. Choose a classification model and train it.
 6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
 7. Fine-tune using GridSearchCV and deploy the best model
6. An agricultural company wants to predict the best crops to grow based on soil composition, rainfall, and past harvest data. What type of machine learning approach would be suitable? Outline the steps to build the model.

A: Supervised Learning – Classification (because the output is a category: best crops to grow).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the data into training and testing sets.
5. Choose a classification model and train it.
6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
7. Fine-tune using GridSearchCV and deploy the best model

7. A business wants to analyze customer responses to different types of advertisements and promotions to determine the most effective marketing strategy. What type of machine learning approach would be suitable? Outline the steps to build the model.

A: Supervised Learning – Classification (because the output is a category: customer response is clicked advertisements and promotions or not).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
 2. Encode categorical using OneHotEncoder or LabelEncoder.
 3. Scale numerical features using StandardScaler.
 4. Split the data into training and testing sets.
 5. Choose a classification model and train it.
 6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
 7. Fine-tune using GridSearchCV and deploy the best model
8. A software company wants to build a system that can analyze source code and predict whether a particular piece of code is likely to contain a bug. What type of machine learning approach would be suitable? Outline the steps to build the model.

A: Supervised Learning – Classification (because the output is a category: code contains bug or not).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
 2. Encode categorical using OneHotEncoder or LabelEncoder.
 3. Scale numerical features using StandardScaler.
 4. Split the data into training and testing sets.
 5. Choose a classification model and train it.
 6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
 7. Fine-tune using GridSearchCV and deploy the best model
9. A fitness app wants to recommend personalized workout plans for users based on their exercise history, fitness level, and preferences. What type of machine learning approach would be suitable? Outline the steps to build the model.

A: Unsupervised Learning – Clustering (because there are no labels, and the goal is to group users)

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Select relevant features and scale the features using StandardScaler.
3. Choose a clustering algorithm.
4. Determine the optimal number of clusters using methods like the Elbow Method or Silhouette Score.
5. Train the model and assign cluster labels to customers
6. Analyze each cluster and use insights to recommend personalized workout plans

10. A social media platform wants to detect fake accounts by analyzing user activity, posting patterns, and interactions with other users. Give steps to Achieve.

A: Supervised Learning – Classification (because the output is a category: predict fake or genuine accounts).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the data into training and testing sets.
5. Choose a classification model and train it.
6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
7. Fine-tune using GridSearchCV and deploy the best model